Forecasting Monthly Consumer Price Index (CPI) in the Philippines

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I. Introduction

The Consumer Price Index (CPI) is the primary measure of inflation in the Philippines. It tracks the weighted average change in prices of a fixed basket of consumer goods and services over time relative to a reference (base) year. The CPI dataset used in this study covers a 14-year period, from June 2011 to March 2025, which consists of 166 complete entries with no missing values. The initial dataset contained two features: the date index and the official consumer price index. For analysis, the date was converted into a proper datetime format and filtered to include only CPI data from January 2012 to December 2024, as the years 2011 and 2025 do not contain a full 12-month period.

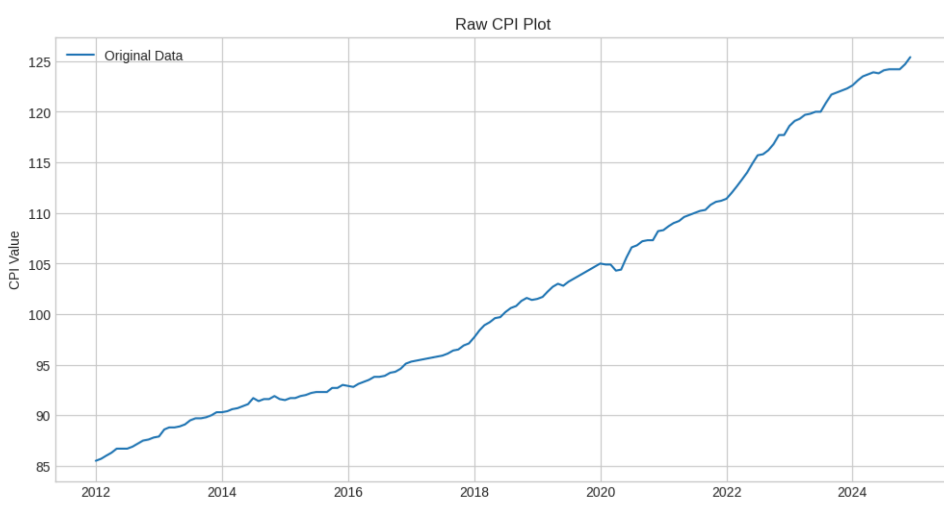
The dataset did not explicitly state the baseline year used for the CPI values. However, this does not significantly affect the analysis since the Philippine Statistics Authority (PSA) only changed the CPI base year in 2025, while the data used in this study mainly covers the period before the change. Therefore, the CPI values from 2012 to 2024 remain comparable and consistent for forecasting purposes.

The initial Exploratory Data Analysis (EDA) revealed that the dataset has a clear upward trend (see Figure 1). The Philippine CPI generally increased over time, which reflects the gradual rise in prices of goods and services driven by inflation and economic activity. While some time series models, such as Autoregressive Integrated Moving Average (ARIMA), require stationary data, more complex approaches do not require such characteristics.

Regional comparisons with four ASEAN economies (Indonesia, Malaysia, Thailand, and Singapore) are also included to provide additional economic context. However, these comparisons are explanatory only, and no forecasting models are developed for countries outside the Philippines.

Figure 1

Raw CPI Plot



The primary objective of this study is to develop an accurate and interpretable forecast of Philippine monthly CPI based on historical data for 2025. In this analysis, two (2) statistical models and four (4) deep learning models are examined and compared, namely, Autoregressive Integrated Moving Average (ARIMA), Seasonal Autoregressive Integrated Moving Average (SARIMA), Recurrent Neural Networks (RNN), Long Short-Term Memory networks (LSTM), 1D Convolutional Neural Networks (1D CNN), and Transformer-based models.

II. Data

A. Source and Coverage

The monthly CPI data used in this study were obtained from the World Bank global inflation database, which compiles inflation indicators for multiple countries and economies. The Philippine headline CPI was selected as the primary variable for analysis because it serves as the standard indicator for inflation and reflects the changes in the general price level of goods and services over time. The use of monthly observations provides sufficient temporal detail for identifying long-term movements and evaluating forecasting performance across statistical and deep learning models

B. Exploratory Data Analysis

The CPI dataset covers a 14-year period, from June 2011 to March 2025. It consists of 166 complete entries with no missing values, indicating that the data is reliable for analysis. However, the dates were stored in YYYYMM format which needed to be converted into proper datetime format to allow accurate time-series analysis and forecasting. The dates were then converted into the proper format, and the dataset was filtered to include only January 2012 to December 2024. The year 2011 was removed because it begins in June and does not contain a full 12-month period, ensuring that only complete yearly data is used for a consistent time-series structure.

In Figure 1, the graph shows a consistent increase in the CPI over the 13-year period. From 2012 to 2020, the trend is relatively linear, indicating steady and predictable growth. However, from 2021 to 2024, the trend becomes much steeper, indicating that the cost of goods and services is increasing at a faster rate compared to earlier years.

A slight dip or plateau is also observed around 2020, likely due to the economic impact of the pandemic, followed by a rapid recovery. Due to this upward trend, differencing or log transformation needs to be performed to make the data stationary before forecasting. It is important to note that in the context of this paper the COVID-19 pandemic spanned from March 11, 2020 to May 05, 2023 based on the World Health Organization’s declarations.

In Table 1 , it shows that the Philippines' CPI growth rate changed noticeably across the three periods. During the pre-COVID period (2012–2020), CPI grew steadily at 2.24 CPI points per year. This sharply increased to 4.89 during the COVID era, reflecting the strong inflationary effects of the pandemic. In the post-COVID period, the growth rate slowed to 3.18, indicating partial economic stabilization while inflation remained higher than pre-pandemic levels.

Table 1

Philippines' CPI Growth Rate by Period

Period	Date Range	Growth Rate (CPI pts/yr)
Pre-COVID	Jan 2012 to Feb 2020	2.24
COVID Era	Mar 2020 to May 2023	4.89
Post-COVID	Jun 2023 to Dec 2024	3.18

Figure 2 compares the CPI trends of the Philippines with selected Southeast Asian countries from 2012 to 2024. Among the countries shown, the Philippines experienced one of the most noticeable increases in CPI, particularly during and after the COVID-19 period highlighted in the graph. While countries such as Thailand and Singapore displayed relatively stable and gradual growth, the Philippines showed a sharper upward trend beginning around 2021, indicating stronger inflationary pressure during the recovery period. Indonesia also exhibited continuous growth, although at a steadier pace, while Malaysia maintained the highest CPI values overall throughout most of the period. Overall, the magnitude and rate of increase differed across economies depending on their economic conditions, recovery strategies, and policy responses.

Figure 2

CPI Comparison Across Different ASEAN Countries

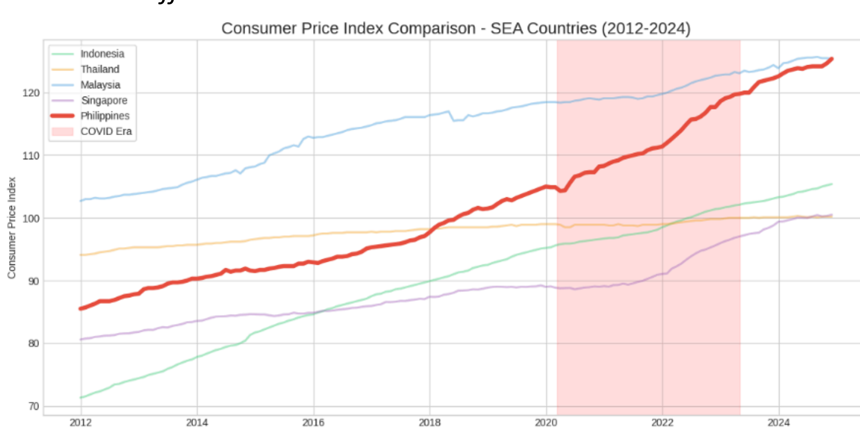


Figure 3 shows both the CPI levels of the Philippines and Singapore from 2012 to 2024 and their year-on-year (YoY) CPI growth rates over the same period. The Philippines exhibited a steeper increase in CPI, particularly during and after the COVID-19 period, indicating stronger inflationary pressure compared to Singapore, which maintained a more gradual and stable rise in consumer prices. Similarly, the YoY growth rates reveal that the Philippines experienced higher and more volatile inflation rates, whereas Singapore generally showed lower and more stable price growth. Overall, the comparison suggests that consumer prices in the Philippines were more sensitive to economic disruptions and recovery periods than those in Singapore.

Figure 3

Comparison of the Philippine CPI Level With a Developed Economy (Singapore)

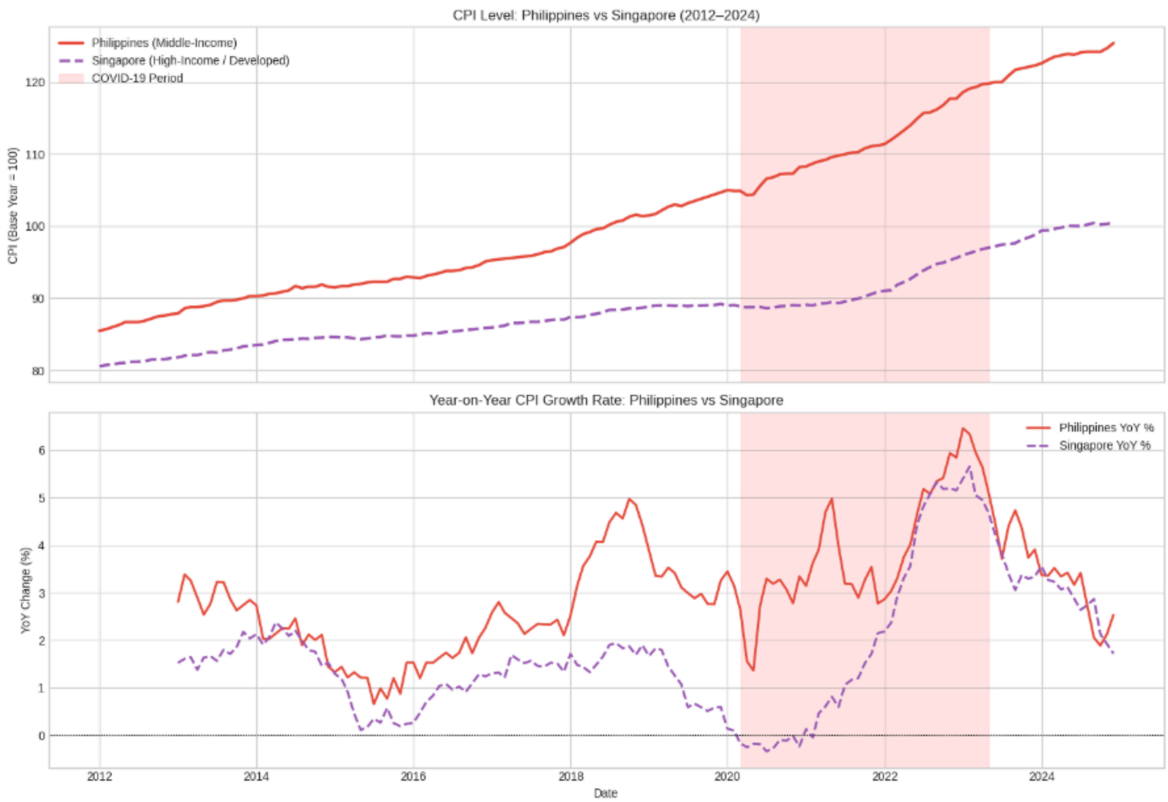
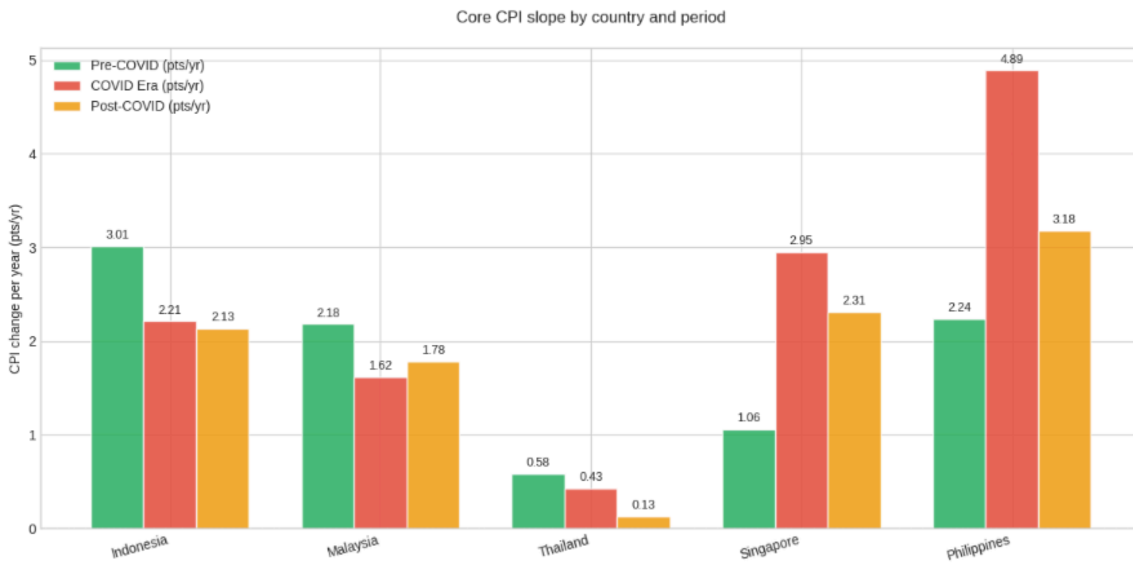


Figure 4 presents the rate of CPI change per year across selected ASEAN countries during the pre-COVID, COVID, and post-COVID periods. Before the pandemic, Indonesia recorded the highest CPI growth rate at 3.01 CPI points per year, followed by the Philippines at 2.24, while Thailand showed the lowest growth at only 0.58. During the COVID era, the Philippines experienced the largest increase among all countries, rising sharply to 4.89 CPI points per year, which reflects the strong inflationary impact of the pandemic on the country. Singapore also showed a significant increase from 1.06 to 2.95, while Indonesia and Malaysia experienced slower growth during the same period. In the post-COVID period, CPI growth generally slowed across all countries, suggesting partial economic stabilization after the pandemic. However, the Philippines still maintained the highest post-COVID growth rate at 3.18 CPI points per year, indicating that inflationary pressures remained relatively high compared to the other ASEAN economies.

Figure 4

The Rate of CPI Change per Year Across Different ASEAN Countries



A multiplicative decomposition was used to decompose the CPI time series into trend, seasonality, and residual components. This approach is the industry standard for CPI and most economic data because inflation is expressed in percentage terms, meaning changes are relative to the current price level (Hyndman & Athanasopoulos, 2018). As CPI increases over time, seasonal effects and variations also scale it, which is better captured by a multiplicative model than an additive one.

Shown in Figure 5 are the decomposition of the monthly raw CPI data from raw, trend, seasonality, and residuals.

Figure 5.1 shows the observed data that indicates that prices consistently increased from 2012 to 2024, with a noticeable acceleration after 2021. The trend component in Figure 5.2 shows steady and almost linear growth from 2012 to 2020. After 2021, an inflection point is evident where the slope becomes steeper, indicating a faster growth rate. In a multiplicative model, this suggests that the underlying percentage increase has accelerated over time. The seasonality component in Figure 5.3 reveals a consistent repeating pattern every 12 months. However, the magnitude of this effect is very small (around ±0.05%), indicating that seasonal changes have minimal impact compared to the overall trend. Finally, the residuals in Figure 5.4 show a significant downward spike around 2020. This represents an unusual event, likely due to the economic impact of the pandemic, where prices were approximately 1% lower than expected based on the trend.

Figure 5

Decomposition of the monthly raw CPI data

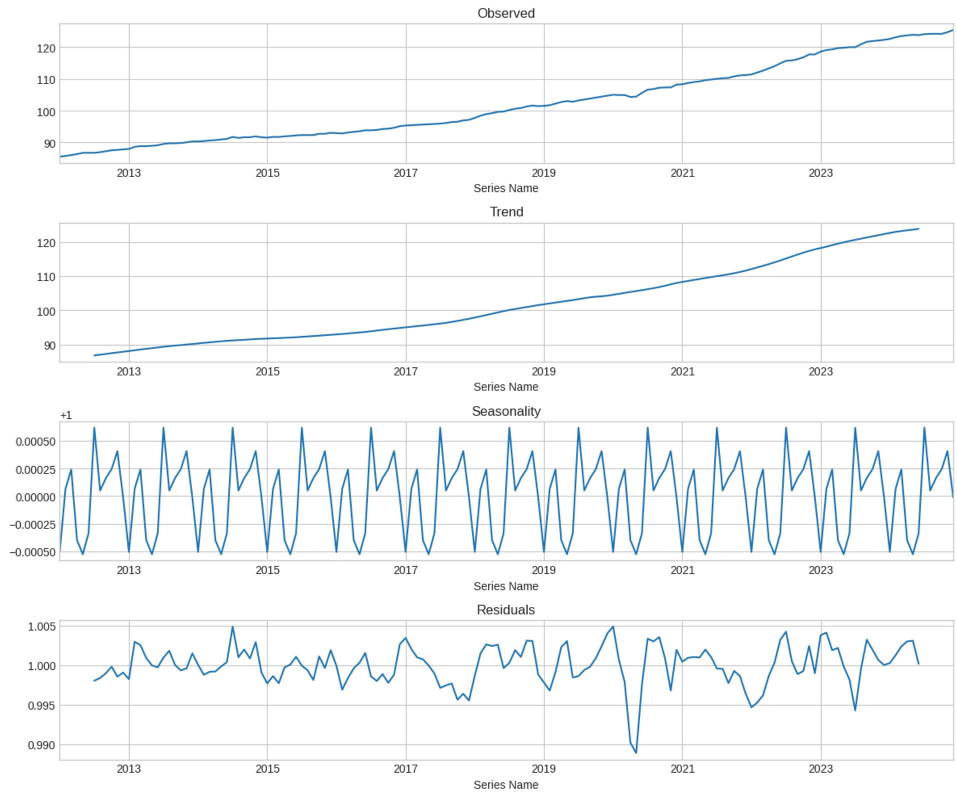


Table 2 shows the results of the Augmented Dickey-Fuller (ADF) test for stationarity indicate that the data is non-stationary. This is supported by the positive ADF statistic (2.70) and the very high p-value (0.999), which suggest the presence of a persistent trend and a changing mean over time. Therefore, differencing or log transformation needs to be performed before the data can be used for accurate forecasting.

Table 2

ADF results of raw CPI data

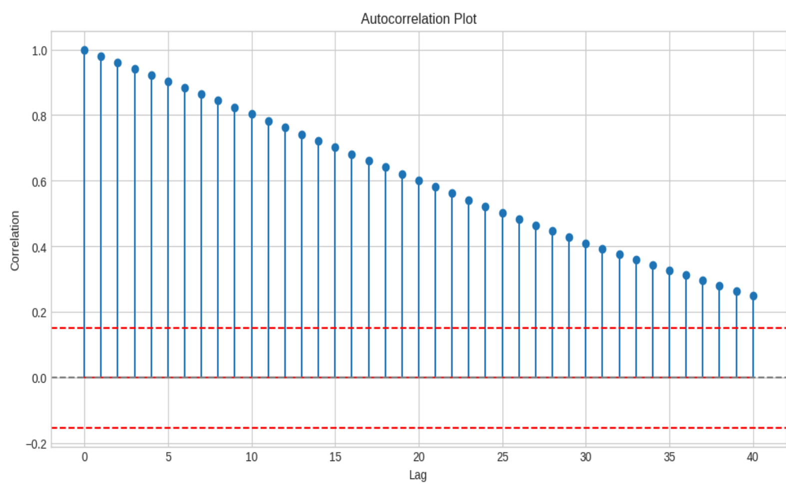
Metrics	Raw Data
ADF Statistic	2.702

p-value	0.999
Status	Non-stationary

The autocorrelation plot in Figure 6 displays the correlation of the CPI across lags up to 40. The correlation starts at 1.0 for lag 0 and gradually declines, reaching around 0.25 by lag 40. The blue stems represent autocorrelation values, while the dashed red lines mark the confidence bounds at approximately ±0.15. Values outside these bounds indicate statistically significant autocorrelations. This pattern suggests a strong persistence at lower lags that weakens over time.

Figure 6

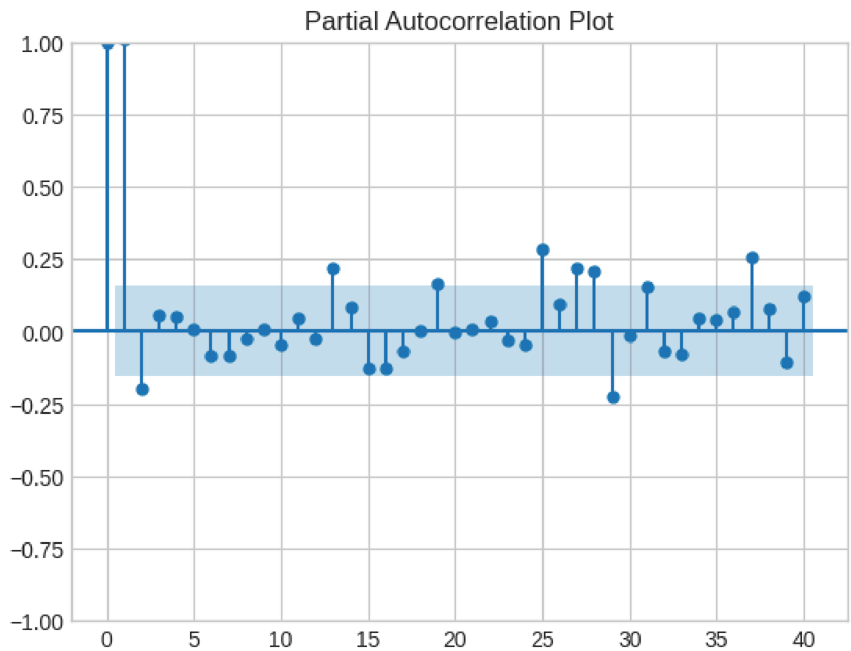
Autocorrelation plot of raw CPI data



The partial autocorrelation results shown in Figure 7 reveal strong persistence in the series, with correlations remaining significant across many lags. The slow decay pattern points to a trend-like or autoregressive process rather than random noise. Since correlations continue to be significant even at longer lags, the series appears non-stationary, indicating that differencing will be required before appropriate time series modeling can be applied.

Figure 7

Partial Autocorrelation plot of raw CPI data

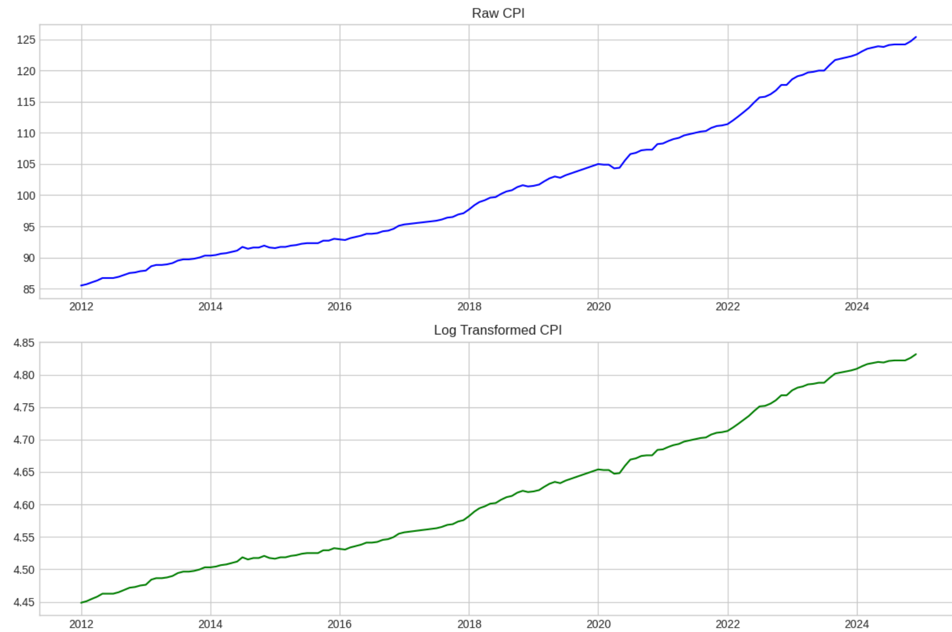


C. Data Preprocessing

As shown in Figure 8, the raw CPI series displays a steady upward trend from 2012 to 2024, reflecting continuous growth in consumer prices. After applying the logarithmic transformation, the series becomes smoother and more linear, which helps stabilize variance and makes long-term growth patterns easier to interpret. This adjustment is particularly useful for preparing the data for stationarity testing and model development, though the persistence of the upward movement suggests that further differencing may still be necessary to achieve a stationary series.

Figure 8

Comparison of Raw CPI and Log Transformed CPI data



The ADF test results after applying the logarithmic transformation, seen in Table 3, confirms that the series remains non-stationary. The ADF statistic of 1.83 and the extremely high p-value of 0.998 indicate that the null hypothesis of non-stationarity cannot be rejected. This means that despite the logarithmic transformation smoothing the growth pattern, the series still exhibits trend or seasonality, and further differencing will be required to achieve stationarity before reliable time series modeling can be performed.

Table 3

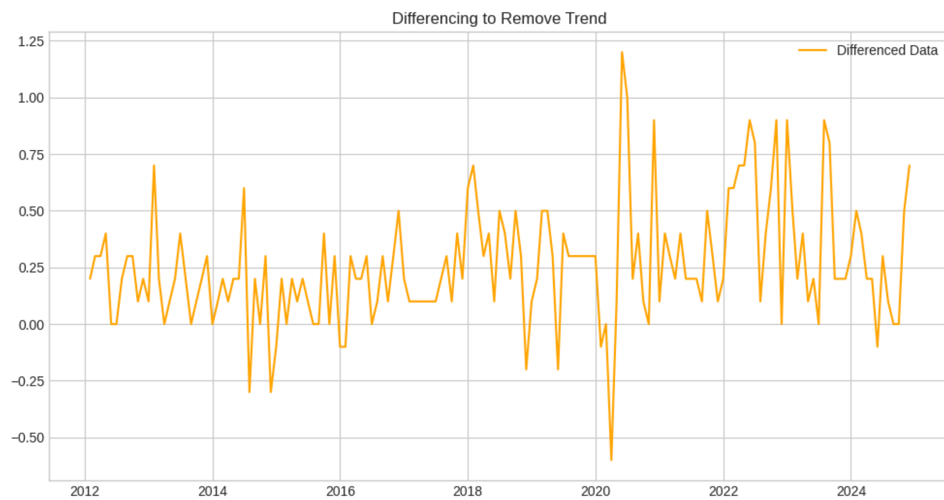
Summarized results of ADF test for both raw and logarithmic transformed data

	Raw Data	**Logarithmic Transformed**
ADF Statistic	2.702	1.834
p-value	0.999	0.998
Status	Non-Stationary	Non-Stationary

As shown in Figure 9, applying first differencing to the CPI series successfully removes the upward trend, producing data that fluctuates around zero. The ADF test confirms this transformation achieves stationarity, with a statistic of -9.39 and a p-value near zero. These results strongly reject the null hypothesis of non-stationarity, indicating that the differenced series is stable and suitable for time series modeling.

Figure 9

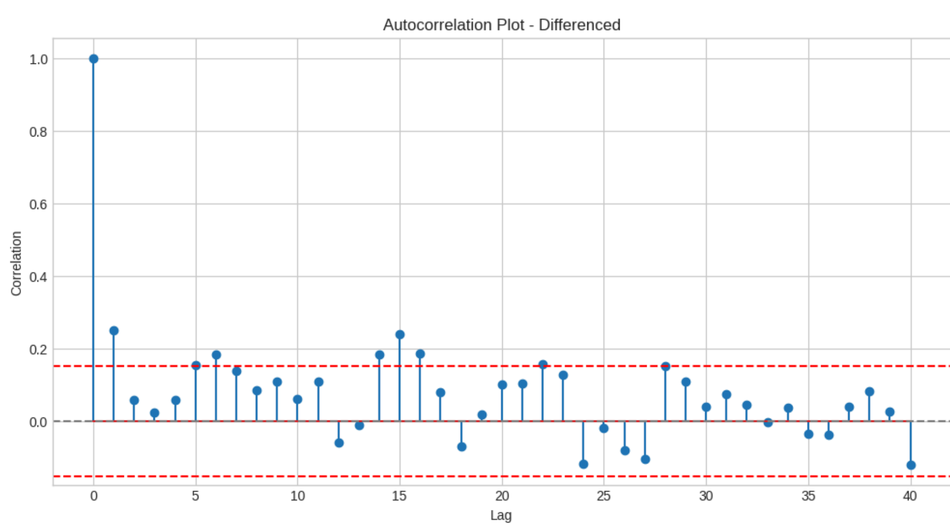
Differenced CPI data



In Figure 10, the differenced CPI series is stationary, but the autocorrelation plot reveals important features. There is a significant spike at lag 1, a negative correlation at lag 12, and strong spikes at lags 15 and 16. The repeating pattern at lags 12, 24, and 36 further suggests seasonality. These insights indicate that while differencing removed the trend, seasonal components may still need to be incorporated into the model.

Figure 10

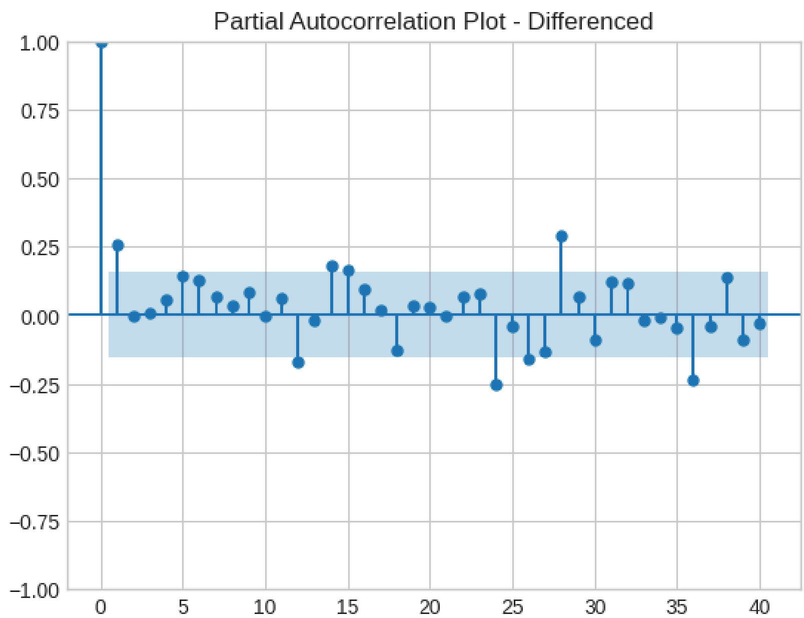
Autocorrelation plot of differenced CPI data



The partial autocorrelation of the differenced CPI series in Figure 11 reveals significant spikes at lags 1 and 2, pointing to short-term autoregressive effects. Seasonal behavior is evident with repeating patterns at lags 12, 24, and 36, though lag 12 is only mildly significant. More importantly, lags 24 and 28 break out of the confidence bounds, reinforcing the presence of seasonality that should be incorporated into the model.

Figure 11

Partial autocorrelation plot of differenced CPI data



III. Methodology

This study employed a quantitative, comparative approach to forecasting the Philippine core CPI using both classical statistical models and modern deep learning architectures. This was implemented in Python, utilizing publicly available monthly CPI data. The overall workflow, from data collection through preprocessing, model training, evaluation, and final forecasting, is systematically illustrated in the conceptual framework seen in Figure 12.

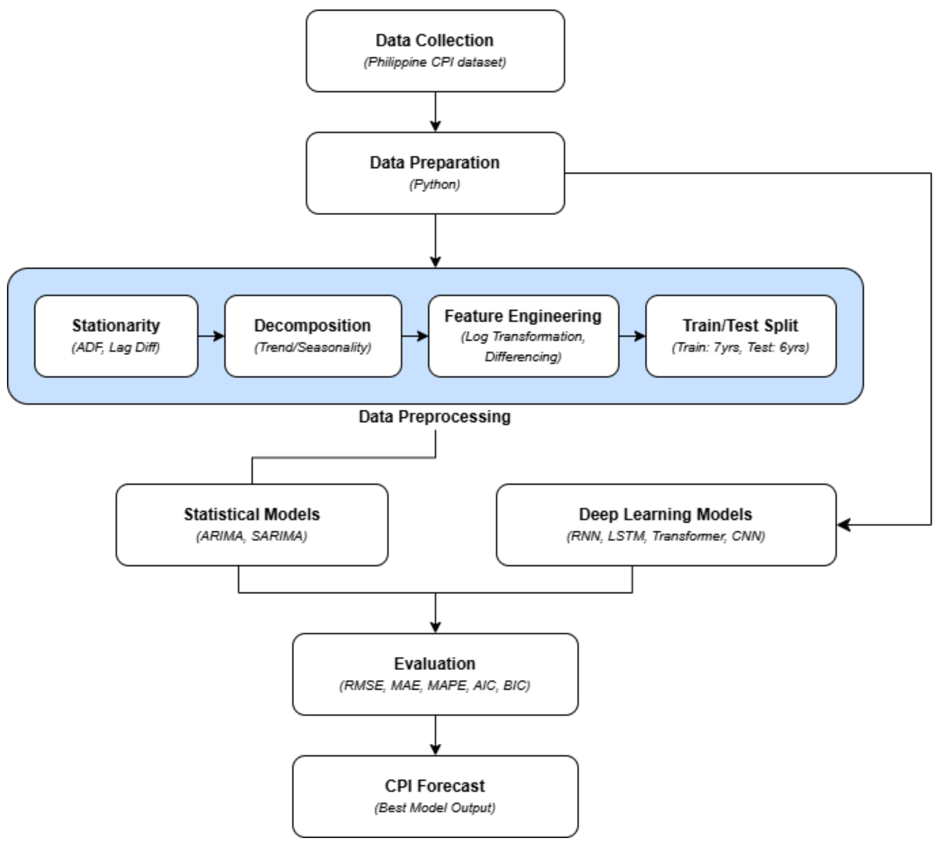
A. Conceptual Framework and Train-test Split

Figure 12 presents the conceptual framework guiding the methodological design of this study. The process begins with the collection of the Philippine CPI dataset, followed by data preparation in Python. The core preprocessing stage encompasses four steps: which are stationarity testing using the ADF test and differencing; time series decomposition to identify underlying trend and seasonality; feature engineering through log transformation and differencing transformations; and the train-test split. From the preprocessed data, two types of forecasting approaches were used: statistical models (ARIMA and SARIMA) and deep learning models (RNN, LSTM, Transformer, and CNN). However, it is important to note that the deep learning models used the filtered dataset and not the stationary dataset as deep learning models do not require stationary data. All models were evaluated against a common set of performance metrics, with the best-performing model used to generate the final CPI forecast.

Regarding the train-test split, the dataset spans from January 2012 through December 2024, comprising 156 monthly observations. The last 6 years were reserved as the test set, leaving 7 years for training. The 6-year test window was deliberately created to be long to test the model in real and varied economic conditions.

Figure 13

Conceptual Model



B. Evaluation Metrics

To properly interpret the results of each model, it is essential to compare it relative to the scale of the data. Shown in Table 4 are the basic statistical results of the dataset. The dataset has a mean CPI of 102.102 and a standard deviation of 11.80, indicating moderate variability over time. The performance of each model was evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). For the statistical models, ARIMA and SARIMA, Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) were also used to measure model fit while accounting for model complexity. To ensure fair comparison across all models, the error metrics were calculated using the original CPI values after applying the necessary log transformation and differencing for ARIMA/SARIMA and inverse MinMax scaling for the deep learning models.

Generally, there is no universal standard for MAE and MAPE in CPI prediction, as both metrics are scale-dependent and vary according to the magnitude of the data being predicted. However, according to Lewis (1982) as cited in Klimberg et al. (2010), highly accurate forecasts are associated with the model if the MAPE is less than or equal to 10%. With this, a 5% error would translate to a forecast deviation of approximately 5.11 CPI index points relative to the dataset mean. Given that the dataset's standard deviation is 11.80, a 5% error represents less than half of the natural variability in the data, indicating that the forecast errors remain within a reasonable and manageable range.

Table 4

Basic Statistical Values of the Dataset

Statistical Values	Result
Mean	102.102
Standard Deviation	11.80
Minimum	85.5
Median	99.95
Maximum	125.4

Furthermore, from a project perspective, this level of accuracy suggests that the model is capable of capturing the general movement and trend of CPI over time, making it sufficiently reliable for economic analysis, planning, and inflation monitoring. While the forecast may not perfectly match every observed value, the relatively small average deviation implies that decisions based on the model's outputs would still be grounded in realistic estimates of price behavior. Therefore, a MAPE of 5% in this context reflects a high level of predictive accuracy and supports the overall reliability of the forecasting model for the intended application.

C. Statistical Models

The first statistical model tested is the Autoregressive Integrated Moving Average (ARIMA). The ARIMA models were fitted on the log-transformed CPI series (requiring first-order differencing, d=1, to achieve stationarity). Two variants were tested:

- ARIMA (1,1,1): An autoregressive order of 1 was used to capture parsimonious short-term dependence, while a moving average order of 1 was selected based on a significant spike at lag 1 in the ACF of the differenced series.

- ARIMA (2,1,1): An autoregressive order of 2 was chosen based on the PACF showing significance at lags 1 and 2, allowing the model to capture additional short-term momentum in CPI growth rates.

Then, Seasonal Autoregressive Integrated Moving Average (SARIMA) extended the ARIMA framework with seasonal components at period S=12 (annual cycle). Seasonal parameters (P=1, D=1, Q=1) account for seasonal autoregression, seasonal differencing to remove the annual cycle, and a seasonal moving average term. Two variants mirroring the ARIMA specifications were also evaluated: SARIMA (1,1,1)(1,1,1,12) and SARIMA (2,1,1)(1,1,1,12).

All statistical model forecasts, reported in log-scale by the fitted model, were back-transformed via exponentiation to the original CPI scale before computing accuracy metrics and generating plots.

D. Deep Learning Models

For the deep learning approach, four deep learning architectures were trained and evaluated, which are RNN, LSTM, a Transformer, and a 1D CNN. These models were trained directly on MinMax-scaled raw CPI values (range [0,1]) using a 12-month lookback window. All models used the same train/test split as the statistical models, with inverse-scaling applied before metric computation to ensure comparability. Moreover, All four deep learning models were trained for 10 epochs with a batch size of 32, using 10% of the training data as a validation split. A fixed random seed of 42 was applied across NumPy, TensorFlow, and Python's random module to ensure reproducibility across runs.

The first deep learning model evaluated is the RNN. The Simple RNN serves as the baseline recurrent model. It consisted of a single recurrent layer with 50 units followed by a dense output layer. While RNNs are capable of capturing short-term temporal dependencies, they are susceptible to the vanishing gradient problem, which may limit their ability to learn patterns across longer time horizons.

Second, the LSTM addresses this limitation through its gating mechanism, which includes input, forget, and output gates. This allows the network to selectively retain or discard information across time steps. The architecture is similar to the RNN, with a single LSTM layer of 50 units and a dense output layer, but the gating mechanism enables it to capture longer-range dependencies within the 12-month lookback window.

Third, the Transformer model applies multi-head self-attention to relate all time steps in the input sequence simultaneously instead of processing them sequentially. The architecture begins with a linear projection into a 64-dimensional embedding space, followed by a Transformer block with four attention heads and a feed-forward dimension of 64. This is followed by global average pooling, a dropout layer with a rate of 0.2, and a dense layer with 25 units using ReLU activation before the output layer. Layer normalization and residual connections are included within the Transformer block to stabilize training.

Finally, the 1D CNN model extracts local temporal features through convolutional filters applied along the time axis. It consists of two consecutive Conv1D layers, each with 64 filters and a kernel size of 3, using ReLU activation. This is also followed by global average pooling, a dense layer with 50 units, and a linear output layer. The convolutional structure makes it effective at detecting repeating short-range patterns, such as consistent month-to-month price movements observed in CPI data.

IV. Results

A. Model Performance

The seasonal component added with the SARIMA models proves to be an essential component with CPI forecasting (see Table 5). In terms of RMSE, MAE, and MAPE, the SARIMA models performed better than the standard ARIMA models. The SARIMA Model (2,1,1) (1,1,1,12) had the best metrics besides AIC/BIC score. Despite the higher AIC/BIC score, its predictive power is substantially better than the ARIMA models that had a lower AIC/BIC score. On average, the model only deviated from the actual CPI data by approximately two (2) units, which is a seven (7) to eight (8) unit difference than the ARIMA models.

Table 5

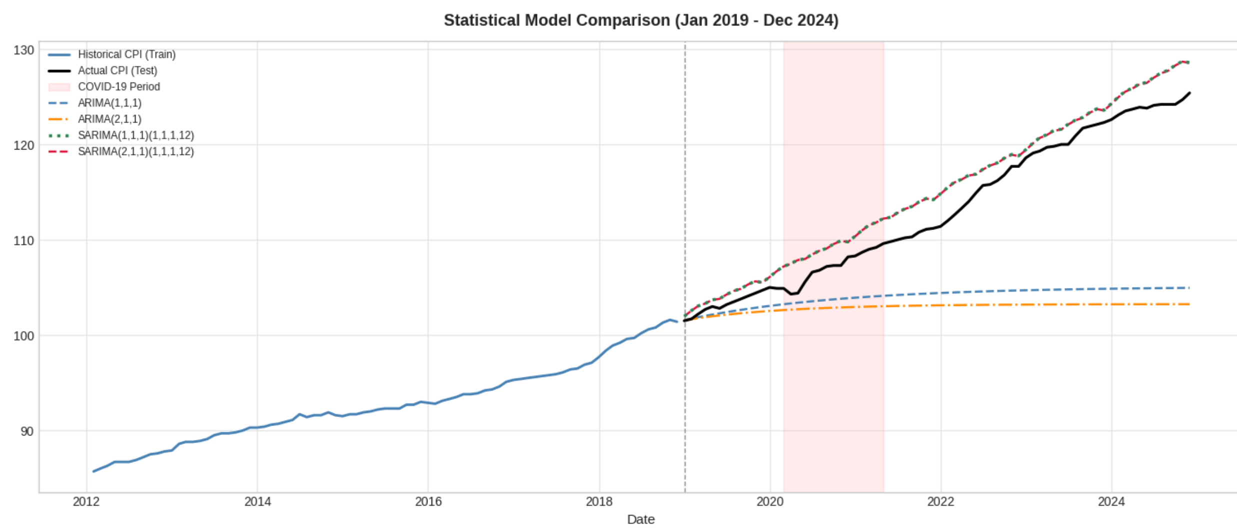
Statistical Models Performance Summary

Metrics	ARIMA (1,1,1)	ARIMA (2,1,1)	SARIMA (1,1,1) (1,1,1,12)	SARIMA (2,1,1) (1,1,1,12)
RMSE	11.260	12.446	2.273	2.249
MAE	9.006	10.132	2.086	2.061
MAPE	7.61%	8.58%	1.84%	1.82%
AIC	-768.02	- 764.49	-615.298	-613.883
BIC	-760.80	-754.86	-604.056	-600.392

As shown in Figure 13, the SARIMA model (2,1,1) (1,1,1,12) closely predicted the actual CPI values which the ARIMA models did not. However, it can be seen that both SARIMA models overpredicted the CPI data which can be attributed to the train-test split. The model was trained on data prior to the COVID-19 pandemic which the model failed to take into account. Despite this, by 2023, the models closely predicts the actual CPI value, and closely mimics the actual trend of the CPI values. The decision between the two SARIMA models ultimately relied on the RMSE, MAE, and MAPE results where the SARIMA model (2,1,1) (1,1,1,12) performed better.

Figure 13

Statistical Models Forecast



With regards to deep learning models, the LSTM and 1D CNN performed the best in terms of RMSE, MAE, and MAPE which can be attributed to the nature of the models. LSTM has a three gating mechanism that can retain or discard information across long sequences. While 1D CNN captures the local momentum using its local temporal features through convolutional filters. With seasonality being the essential component with the seasonal models, having a model that can capture seasonality is important. In contrast, models like the simple RNN do not have a gating system or local temporal features which fail to consider longer trends. Moreover, the transformer model failed as it is a model that requires numerous observations to account for the relationship between every time step and every other time step at the same time.

Table 6

Deep Learning Models Performance Summary

Metrics	RNN	LSTM	Transformer	1D CNN
RMSE	10.879	6.347	24.025	6.678
MAE	9.775	5.753	23.149	6.094
MAPE	8.41%	4.95%	20.19%	5.25%

Among the evaluated models, the SARIMA model (2,1,1) (1,1,1,12) demonstrated the best overall performance as it achieved a RMSE of 2.249, MAE of 2.061, MAPE of 1.82%. Although the other SARIMA model had similar results, the chosen model performed better in terms of the aforementioned metrics. Despite the deep learning models (LSTM and 1D CNN) showing competitive results, there is no improvement in accuracy that can outweigh the resource requirements of deep learning models. Simpler models like SARIMA already delivered reasonably accurate forecasts with lower computational demand and greater interpretability, making SARIMA the more practical choices for CPI forecasting applications.

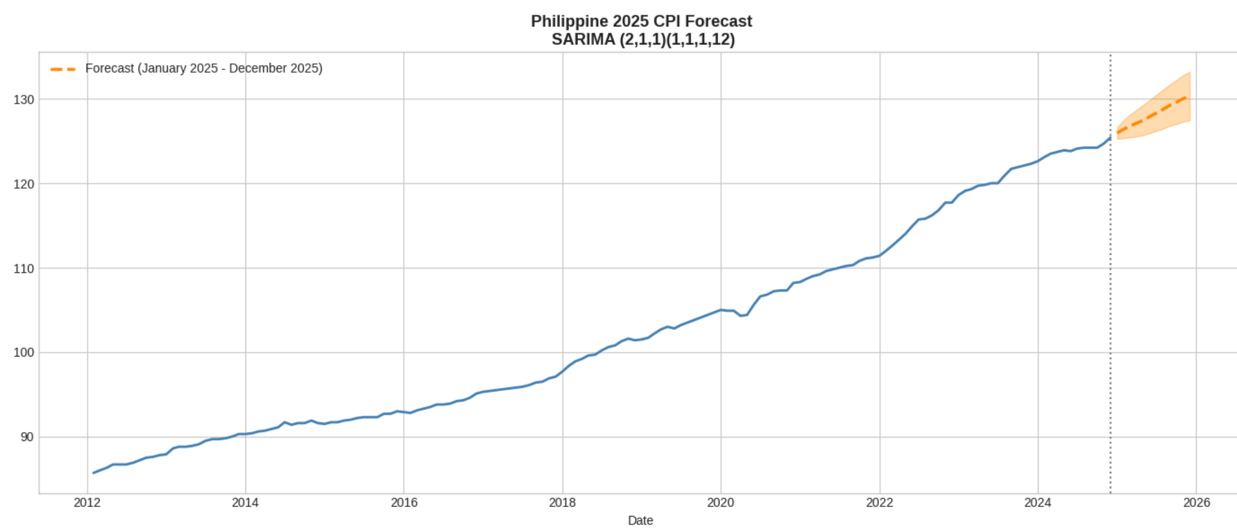
Lastly, to rule out overfitting or underfitting, in-sample performances were evaluated using the same methodology. The chosen SARIMA (2,1,1) (1,1,1,12) model exhibited only a +0.97 percentage-point gap between in-sample and out-of-sample MAPE. This indicates that the model fits its training data and generalizes well to unseen data.

B. Forward Forecast

Using the SARIMA model (2,1,1) (1,1,1,12), the forward forecast for 2025 was conducted. As shown in Figure 14, the forecast projects a continued CPI growth. The widening of the interval indicates a growing uncertainty as the forecast length increases.

Figure 14

2025 Forecast



V. Discussion

The chosen SARIMA model exhibits overprediction during the test period which can be attributed to the training set not including the COVID-19 pandemic. Despite this, the prediction closely mimics the actual CPI data and achieves a MAPE of 1.82% which Lewis (1982) classifies as highly accurate. The model also performed well in terms of RMSE and MAE, with 2.249 and 2.061 respectively. This indicates that the forecast deviation remains small with consistency.

Despite the accurate forecast of the model, actual Philippine CPI value significantly decreased in 2025. According to the Department of Finance, this can be attributed to policy changes such as lowering the maximum suggested retail price (MSRP) which such an intervention could never have been anticipated by any model. The sudden stabilization of the economy during 2025 can be considered as to why PSA will change its baseline year, as it is the most recent year where the economy was considered stable. However, the CPI value rose again with an inflation rate of 7.2% in April 2026 due to worldwide issues like the war in the Middle East (Cigaral & De Leon, 2026). This showcases that SARIMA models, although accurate, are only valid within the assumption that historical data continue to follow a predictable pattern. Human interventions could never be predicted but nevertheless, the prediction can serve as a "what-if" scenario if such an intervention did not occur.

VI. Conclusion

The study successfully developed and evaluated multiple forecasting models for predicting the monthly Consumer Price Index (CPI) in the Philippines using the historical data from 2012 to 2024. Through the exploratory data analysis, stationary testing, and model comparison, the results showed that CPI data contains a strong upward trend with seasonal patterns that significantly affect forecasting performance. Among the models tested, the SARIMA (2,1,1) (1,1,1,12) model produced the best overall results, achieving the lowest RMSE, MAE, and MAPE values. The values also revealed that incorporating seasonality is essential in CPI forecasting, as seasonal models consistently outperformed standard ARIMA models. Although deep learning models such as LSTM and 1D CNN showed competitive performance, the SARIMA model remained more practical due to its lower complexity, interpretability, and strong predictive accuracy.

Overall, the study demonstrates that SARIMA is an effective and reliable approach for forecasting Philippine CPI under the normal economic conditions. While the model showed slight overprediction during periods influenced by external events such as the COVID-19 pandemic and government policy interventions, it was still able to closely capture the long-term movement and trend of inflation. This highlights both the usefulness and limitations of time series forecasting, as models rely heavily on historical patterns and cannot fully anticipate sudden economic disruptions. Nevertheless, the forecasts generated in this can still serve as a valuable tool for inflation monitoring, economic planning, and policy analysis. Future studies may improve forecasting performance by incorporating external economic variables and exploring hybrid forecasting approaches that combine statistical and deep learning methods.

VII. References

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